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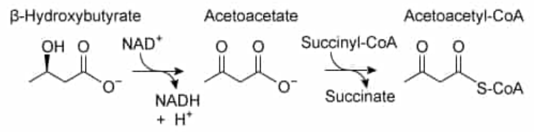

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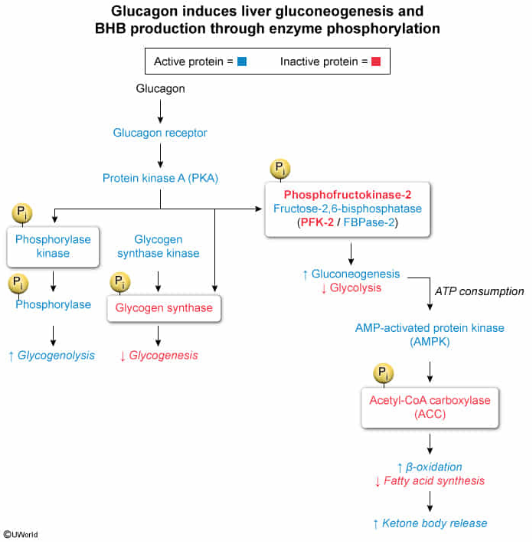
Based on the passage, hepatocytes (ie, liver cells) that are producing BHB are also most likely to have:

- A. downregulated phosphofructokinase-2 (PFK2) activity. [41%]
- B. downregulated phosphorylase kinase activity. [12%]
- C. upregulated acetyl-CoA carboxylase (ACC) activity. [36%]
- D. upregulated glycogen synthase activity. [8%]

Correct

41% answered correctly

Explanation:



In living systems, the regulation of various pathways must happen in a **coordinated manner** to ensure that only the pathways that support the organism's current needs are active at any given time. A **hormone** acts as a common signal that coordinates activity across several organs and tissues. Once a hormone reaches a cell within a target tissue, the hormone can activate tissue-specific enzymes and pathways. One of the ways that hormones and other signals activate biochemical pathways is through posttranslational modifications (eg, **phosphorylation**).

The question asks about the activity of enzymes in hepatocytes (ie, liver cells) that are producing BHB. The liver produces ketone bodies like BHB under glucose-poor (ie, fasting) conditions. During fasting, **glucagon** stimulates the liver to **produce and export metabolic fuels**, which include both BHB (through β-oxidation) and glucose (through **glycogenolysis** and **gluconeogenesis**).

To regulate the glycolysis–gluconeogenesis equilibrium, glucagon regulates the activity of the bifunctional enzyme **phosphofructokinase-2/fructose-2,6-bisphosphatase (PFK-2/FBPase-2)**. This enzyme regulates the level of fructose 2,6-bisphosphate (F2,6BP), which is an allosteric stimulator of glycolysis. Under fasting conditions in which BHB is also being produced, liver cells use gluconeogenesis to provide fuel to other cells, so glycolysis in these cells would be detrimental to the organism overall. Therefore, **phosphofructokinase-2 activity is most likely downregulated** in these cells to *downregulate glycolysis*.

(Choice B) In liver cells, glucagon activates the glucagon receptor, a $G_{s\alpha}$-coupled GPCR. This eventually leads to the activation of protein kinase A (PKA). PKA phosphorylates phosphorylase kinase, thereby *upregulating* its activity to stimulate glycogenolysis.

(Choice C) Acetyl-CoA carboxylase produces malonyl-CoA, which would prevent β-oxidation and BHB production. Therefore, fasting states result in PKA and AMP-dependent protein kinase (AMPK) both phosphorylating and *inhibiting* acetyl-CoA carboxylase (ACC) (ie, *downregulating* ACC activity).

(Choice D) In the fasting state, glycogen synthesis is inhibited; therefore, glycogen synthase activity is expected to be *downregulated*. Glycogen synthase is active in the *fed state*, stimulated by insulin.

Educational objective:
Ketone bodies are produced under fasting conditions. In the fasting state, glucagon signals the phosphorylation and regulation of various enzymes to increase the production of glucose through

both glycogenolysis and gluconeogenesis.

Time spent: 0 sec	**QID**:404126
Subject: Biochemistry	**System:** Metabolic Reactions

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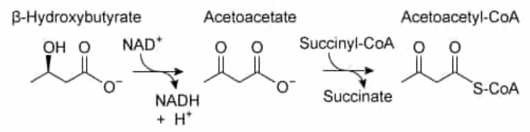

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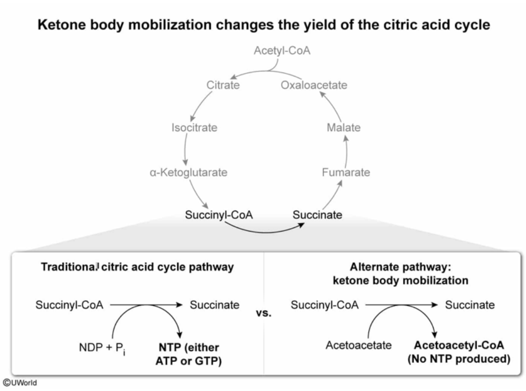
The second reaction shown in Figure 1 involves an enzyme that transfers coenzyme A from succinyl-CoA to acetoacetate. If this enzyme were used in the citric acid cycle in place of succinyl-CoA synthetase, the overall yield of high-energy molecules in one round of the citric acid cycle would be:

- A. decreased by one NTP (either ATP or GTP). [52%]
- B. decreased by one NADH. [18%]
- C. decreased by one $FADH_2$. [12%]
- D. unchanged. [16%]

Correct
53% answered correctly

Explanation:



A biochemical pathway is generally a series of enzyme-catalyzed reactions that convert a reactant into an ending product through a series of metabolic intermediates (metabolites). Many biochemical pathways **share metabolites**, providing opportunities for molecules to leave or reenter an otherwise well-defined pathway. By proceeding through these alternate pathways, the yield (or consumption) of high-energy molecules may change.

The question asks for the change in yield of the citric acid cycle (TCA cycle) if the succinyl-CoA → succinate reaction shown in Figure 1 replaced the traditional succinyl-CoA synthetase reaction. In the *traditional* succinyl-CoA synthetase–catalyzed reaction, the free energy released after hydrolyzing the high-energy thioester bond of succinyl-CoA is used to **form an NTP** (either ATP or GTP, depending on the organism) by **substrate-level phosphorylation**. (Note: This reaction is the **reverse** of a ligase reaction).

The *alternative* pathway shown in Figure 1 is a transfer reaction. The coenzyme A molecule is simply transferred from succinyl-CoA to acetoacetate **without consumption or production of an NTP**. Because no NTP is made in the alternate pathway, but an NTP *is* made in the traditional pathway, the yield of that round of the TCA cycle is **decreased by one NTP molecule**.

(Choices B and C) Although BHB does require NAD^+ to be converted into acetoacetate, the question focuses only on the substitution of the succinyl-CoA → succinate reaction. No redox cofactors (neither NADH nor $FADH_2$) are used in this reaction, nor will their yield be affected by the substitution of this single reaction.

(Choice D) The succinyl-CoA → succinate reaction of the traditional TCA cycle produces an NTP molecule, whereas the succinyl-CoA → succinate reaction shown in Figure 1 does not. Substituting these reactions for each other will therefore alter the NTP yield.

Educational objective:
In the citric acid cycle, the conversion of succinyl-CoA to succinate by succinyl-CoA synthetase is accompanied by the production of GTP (or ATP) by substrate-level phosphorylation. If succinyl-CoA is converted to succinate by an alternative enzyme or pathway, the NTP yield of that round of the citric acid cycle will change.

Time spent:0 sec
Subject:
Biochemistry

QID:404127
System:
Metabolic Reactions

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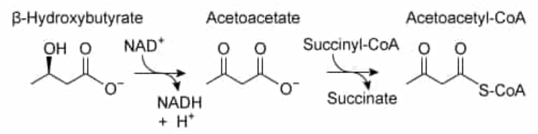

Figure 1 Conversion of BHB to acetoacetyl-CoA

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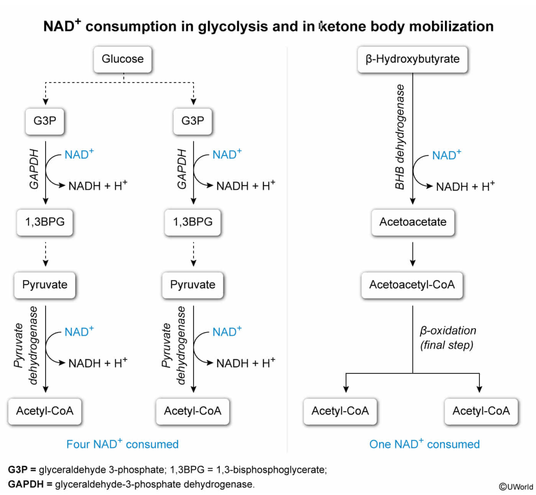
Glucose and BHB can each be converted to two acetyl-CoA molecules. How many NAD^+ molecules (both cytosolic and mitochondrial) are consumed by conversion of glucose to two acetyl-CoA and by conversion of BHB to two acetyl-CoA, respectively?

- A. Two NAD^+ consumed per glucose, one NAD^+ consumed per BHB [38%]
- B. Two NAD^+ consumed per glucose, two NAD^+ consumed per BHB [19%]
- ✔ C. Four NAD^+ consumed per glucose, one NAD^+ consumed per BHB [18%]
- D. Four NAD^+ consumed per glucose, two NAD^+ consumed per BHB [23%]

Correct
19% answered correctly

Explanation:



Nicotinamide adenine dinucleotide (NAD^+) is a commonly used oxidizing agent in biochemistry. As such, many energy-yielding metabolic pathways consume NAD^+ to produce NADH, which can subsequently deposit its high-energy electrons into the electron transport chain to produce ATP by oxidative phosphorylation.

The catabolism of glucose begins in the cytosol with glycolysis. During the energy-investment phase of glycolysis, a single molecule of glucose is broken into **two molecules** of glyceraldehyde 3-phosphate (G3P). In the first step of the energy-payoff phase, the enzyme glyceraldehyde-3-phosphate dehydrogenase (GAPDH) oxidatively phosphorylates G3P to form 1,3-bisphosphoglycerate while also **consuming NAD^+** to produce an NADH. Because **two molecules** of G3P are made per glucose, **two molecules of NAD^+** are consumed by GAPDH per glucose molecule. Glycolysis then continues, ultimately producing **two pyruvates**.

To produce acetyl-CoA, the pyruvates are both oxidatively decarboxylated in the mitochondria by the pyruvate dehydrogenase complex. This enzyme consumes one NAD^+ per pyruvate, and therefore **two NAD^+ are consumed per glucose** at this step. Altogether, producing two acetyl-CoA from glucose requires consumption of **four NAD^+ molecules**.

Figure 1 shows that only one NAD^+ is required in the conversion of BHB to acetoacetyl-CoA. The passage states that acetoacetyl-CoA is then processed through β-oxidation. Although β-oxidation overall consumes NAD^+ at various points, acetoacetyl-CoA is already oxidized. Therefore, when acetoacetyl-CoA is the starting material in β-oxidation, *only* the final step (ie, cleavage with incoming CoA-SH) is required to produce two acetyl-CoA molecules. This step consumes a free coenzyme A but *does not require any additional NAD^+*. Therefore, only **one molecule of NAD^+** is needed to convert BHB into two acetyl-CoA.

(Choices A and B) Although glycolysis consumes two NAD^+ to produce two NADH and two *pyruvates*, an extra two NAD^+ are required to convert the two pyruvates into acetyl-CoA.

(Choice D) Only one NAD^+ is needed to reduce BHB to acetoacetate. The remaining steps to produce two acetyl-CoA do not require NAD^+.

Educational objective:

Glycolysis consumes two NAD^+ to produce two NADH and two pyruvates; an additional two

NAD^+ are required to convert the pyruvates to acetyl-CoA. The conversion of BHB to two acetyl-CoA requires only one NAD^+.

Time spent: 0 sec	**QID**:404128
Subject: Biochemistry	**System:** Metabolic Reactions

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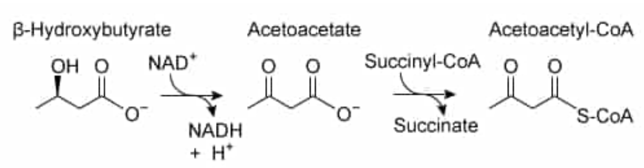

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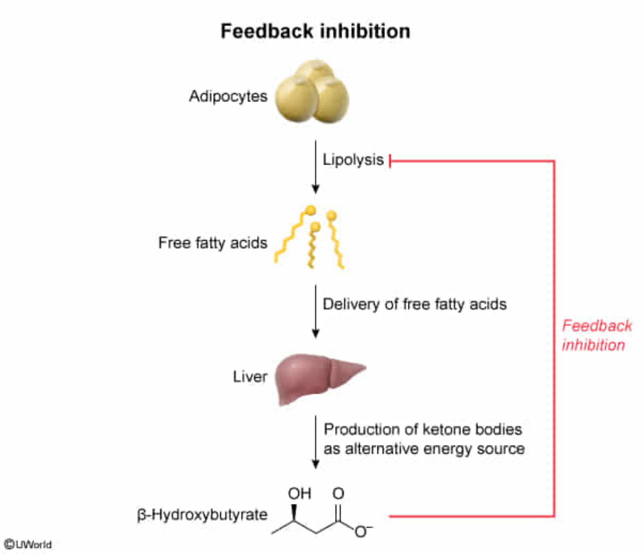
The effect of BHB on adipocyte lipolysis is an example of:

- A. uncompetitive inhibition. [14%]
- B. cooperativity. [5%]
- C. positive feedback. [8%]
- D. negative feedback. [71%]

Correct
71% answered correctly

Explanation:



Biochemical pathways must be regulated to ensure that a given pathway is not over- or under-active for an organism's current needs. One of the most common forms of regulation is feedback regulation, in which a product of a pathway feeds **backward** to an **earlier step** and either stimulates that step (a positive feedback loop that increases levels of the product) or inhibits that step (a **negative** feedback loop that limits the levels of the product).

The ketone body BHB is produced from fat oxidation in the liver. The passage states that BHB **inhibits** adipocyte lipolysis. Lipolysis of adipose tissue (ie, fat tissue) results in the release of free fatty acids into the bloodstream. These fatty acids are taken up by various organs and tissues, including the liver. In the liver, these free fatty acids can be used to **produce the ketone body BHB**.

BHB's **inhibition** of adipocyte lipolysis is an inhibition of an **earlier step in its own synthetic pathway**. Therefore, its inhibition will regulate and **limit** subsequent production of BHB. Consequently, the inhibition of adipocyte lipolysis by BHB is an example of **negative feedback inhibition**.

(Choice A) Uncompetitive inhibition is the inhibition of an enzyme (or receptor) following the inhibitor binding the enzyme–substrate complex (or the receptor–agonist complex). The passage states that BHB acts as an agonist to *activate*, not inhibit, HCAR2, which *subsequently* inhibits lipolysis. It is not stated that BHB binds or inhibits any subsequent enzyme (eg, hormone-sensitive lipase) in an uncompetitive manner.

(Choice B) Cooperativity occurs when one binding event alters the affinity of subsequent binding events. The passage does not describe the binding affinity of BHB on any component involved in adipocyte lipolysis.

(Choice C) If BHB regulated adipocyte lipolysis through a positive feedback loop, it would be expected to *stimulate* lipolysis, not inhibit it.

Educational objective:
Negative feedback is a common mechanism for regulation of biochemical pathways. In negative feedback, the product of a biochemical pathway feeds backward and inhibits an earlier step of the product's synthesis pathway.

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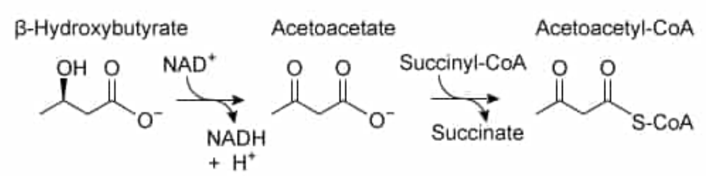

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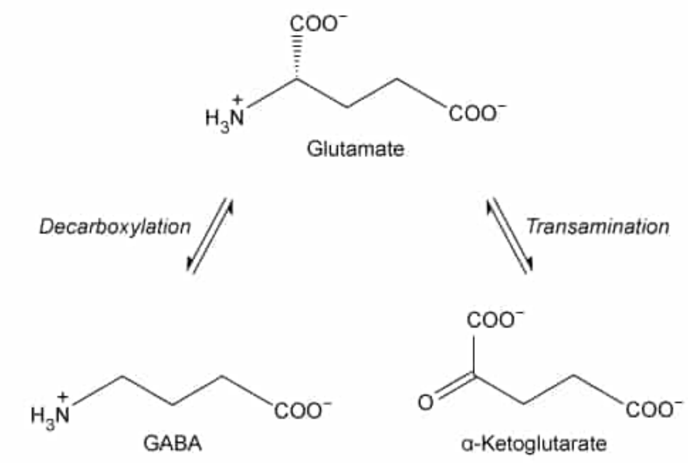
In GABAergic neurons, glutamate can be converted either to GABA or to

α-ketoglutarate.

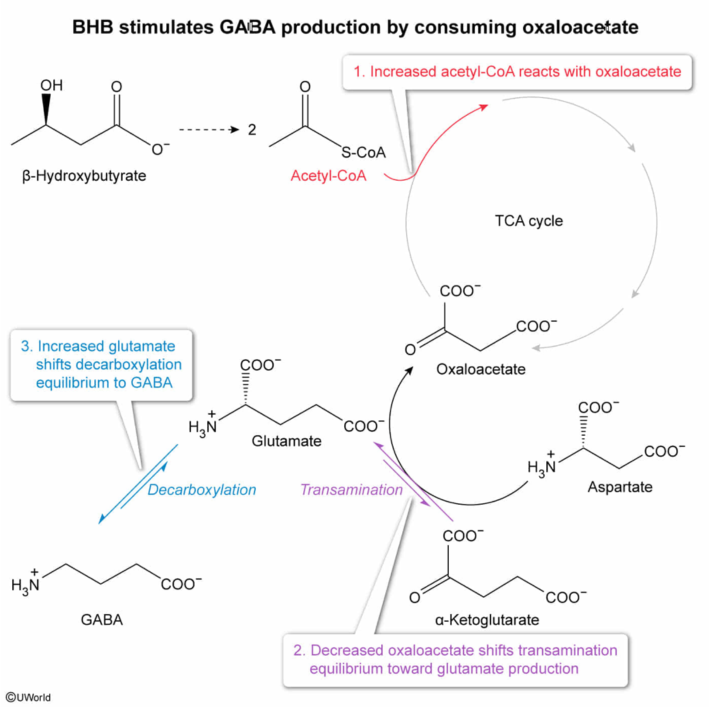


Which statement would best explain the effect of BHB on this equilibrium?

- A. Acetoacetate, produced from BHB oxidation, is the α-keto derivative of GABA and competitively inhibits transamination of α-ketoglutarate. [47%]
- B. BHB is a gluconeogenic substrate that is metabolized into pyruvate, providing an additional α-keto acid substrate for the transamination reaction. [11%]
- C. BHB acts as an allosteric inhibitor of glutamate decarboxylase, the enzyme that catalyzes the decarboxylation reaction. [14%]
- D. Acetyl-CoA produced from BHB metabolism temporarily depletes oxaloacetate, which acts as a cosubstrate for the transamination reaction. [26%]

Correct
26% answered correctly

Explanation:



Many biochemical pathways are interconnected through **shared**

metabolic intermediates (ie, metabolites). If the amount or concentration of any metabolite is perturbed, it can affect the equilibrium —and therefore the metabolite levels—of other reactions within the pathway due to Le Châtelier's principle. Similarly, other pathways may also be affected through **shared intermediates**, even if the shared intermediates were not directly affected by the initial perturbation.

The question shows the equilibrium involving glutamate, GABA, and α-ketoglutarate. The passage states that treatment of GABAergic neurons (neurons that produce and release GABA as a neurotransmitter) with BHB stimulates an increase of GABA synthesis. BHB is a ketone body and, as such, is broken down into acetyl-CoA in the mitochondria.

In mitochondria, acetyl-CoA readily reacts with oxaloacetate during the first step of the citric acid cycle. Increased acetyl-CoA levels temporarily cause this reaction's equilibrium to **shift to the right**, resulting in **increased consumption (ie, decreased levels)** of oxaloacetate. Note that although oxaloacetate levels may eventually be restored upon completion of the cycle, until oxaloacetate levels are restored, the equilibria of downstream reactions will also be perturbed.

Oxaloacetate is an α-keto acid, which can serve as a substrate for transamination reactions, (eg, the conversion of oxaloacetate into aspartate by removing an amine from glutamate). The temporary decrease in oxaloacetate (ie, reactant) causes the glutamate transamination equilibrium shown in the question to **shift to the left to produce more glutamate**.

With glutamate levels increased, glutamate's decarboxylation equilibrium then shifts **toward GABA production**. Therefore, **BHB stimulates GABA production** by **temporarily decreasing oxaloacetate levels** and shifting transamination equilibrium.

(Choice A) Acetoacetate is a β-keto acid, and is not the α-keto derivative of GABA.

(Choice B) BHB is ketogenic, not glucogenic; it **cannot be used for net production of glucose or any glucogenic precursor** (eg, pyruvate).

(Choice C) If BHB inhibited glutamate decarboxylase, then GABA production would be *inhibited* by BHB, not stimulated.

Educational objective:
Many biochemical pathways contain multiple reactions at equilibrium and involve metabolites that are shared between pathways. If the levels of a given molecule in one pathway are perturbed, it can affect many other pathways through the shifted equilibrium of the shared metabolites.

Time spent:0 sec
Subject:
Biochemistry

QID:404130
System:
Metabolic Reactions